

Bash Script : Lab 2

1. Create a script that asks for user name then send a greeting to him.

```
#!/usr/bin/bash
read -p "What's Your name: " name
echo "Hi ${name}"
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh
What's Your name: nagy
Hi nagy
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
```

2. Create a script called s1 that calls another script s2 where in s1 there is a variable called x, it's value 5

```
#script1.sh
#!/usr/bin/env bash
export x=5
./script2.sh
```

```
#script2.sh
#!/usr/bin/env bash
echo "x from environment = $x"
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script1.sh
x from environment = 5
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
```

3. Create a script called mycp where, It copies a file to another, It copies multiple files to a directory

```
#!/usr/bin/bash
#!/usr/bin/env bash

if [ $# -lt 2 ]; then
```

```

    echo "Usage:"
    echo "  ./mycp source_file destination_file"
    echo "  ./mycp file1 file2 ... directory"
    exit 1
fi

if [ $# -eq 2 ]; then
    if [ -f "$1" ]; then
        cp "$1" "$2"
    else
        echo "Error: $1 is not a file"
    fi
    exit 0
fi

dest="${@: -1}"
if [ ! -d "$dest" ]; then
    echo "Error: Last argument must be a directory"
    exit 1
fi

for file in "${@:1:$#-1}"; do
    if [ -f "$file" ]; then
        cp "$file" "$dest"
    else
        echo "Warning: $file is not a file"
    fi
done

```

```

nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script1.sh
Usage:
  ./mycp source_file destination_file
  ./mycp file1 file2 ... directory
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script1.sh script2.sh ./hello.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ls
hello.sh  Lab2.md  script1.sh  script2.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$

```

4. Create a script called mycd where, It changed directory to the user home directory, if it is called without arguments, Otherwise, it change directory to the given directory

```

#!/usr/bin/env bash
if [ $# -eq 0 ]; then
    cd "$HOME" || exit
else
    cd "$1" || echo "Directory does not exist"
fi

```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ source ./script.sh
nagy@ZenBook-UX425EA:~$
nagy@ZenBook-UX425EA:~$ cd -
/home/nagy/Mine/ITI/014_BashScript/Solutions/Lab2
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ source ./script.sh ../
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions$
```

5. Create a script called myls where, It lists the current directory, if it is called without arguments, Otherwise, it lists the given directory

```
#!/usr/bin/env bash
if [ $# -eq 0 ]; then ls
else ls "$1"
fi
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh
Lab2.md script.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh -F
Lab2.md script.sh*
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh ~/Desktop/
'Data Structure Exam2.mhtml' Dev ecommerce_back newusers test_nagy
'Data Structure Exam.mhtml' e-commerce nagy test
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
```

6. Enhance the above script to support the following options individually:

- -l: list in long format
- -a: list all entries including the hiding files.
- -d: if an argument is a directory, list only its name
- -i: print inode number
- -R: recursively list subdirectories

```
#!/usr/bin/env bash

options=""
dir=""

for arg in "$@"; do
    if [ "$arg" = "-l" ]; then options="$options -l"
    elif [ "$arg" = "-a" ]; then options="$options -a"
    elif [ "$arg" = "-d" ]; then options="$options -d"
    elif [ "$arg" = "-i" ]; then options="$options -i"

```

```

    elif [ "$arg" = "-R" ]; then options="$options -R"
    else dir="$arg"
    fi
done

if [ -z "$dir" ]; then
    ls $options
else
    ls $options "$dir"
fi

```

```

nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh -l
total 8
-rw-rw-r-- 1 nagy nagy 3767 Feb 23 16:45 Lab2.md
-rwxrwxr-x 1 nagy nagy 437 Feb 23 16:47 script.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh -la
total 16
drwxrwxr-x 2 nagy nagy 4096 Feb 23 16:37 .
drwxrwxr-x 4 nagy nagy 4096 Feb 23 15:56 ..
-rw-rw-r-- 1 nagy nagy 3767 Feb 23 16:45 Lab2.md
-rwxrwxr-x 1 nagy nagy 437 Feb 23 16:47 script.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh -i
423850 Lab2.md 424318 script.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh -R
.:
Lab2.md script.sh
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$

```

7. Create a script called mytest where, It check the type of the given argument (file/directory), It check the permissions of the given argument (read/write/execute)

```

#!/usr/bin/env bash

if [ $# -eq 0 ]; then
    echo "Usage: ./mytest <file_or_directory>"
    exit 1
fi

item="$1"
if [ -f "$item" ]; then echo "$item is a regular file"
elif [ -d "$item" ]; then echo "$item is a directory"
else echo "$item is not a regular file or directory"
fi

[ -r "$item" ] && echo "$item is readable" || echo "$item is not readable"
[ -w "$item" ] && echo "$item is writable" || echo "$item is not writable"
[ -x "$item" ] && echo "$item is executable" || echo "$item is not
executable"

```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ll Lab2.md  
-rw-rw-r-- 1 nagy nagy 3691 Feb 23 16:51 Lab2.md  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh ./Lab2.md  
./Lab2.md is a regular file  
./Lab2.md is readable  
./Lab2.md is writable  
./Lab2.md is not executable  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh ./script.sh  
./script.sh is a regular file  
./script.sh is readable  
./script.sh is writable  
./script.sh is executable  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$
```

8. Create a script called myinfo where, It asks the user about his/her logname, It print full info about files and directories in his/her home directory, Copy his/her files and directories as much as you can in /tmp directory, Gets his current processes status.

```
#!/usr/bin/env bash  
  
read -p "Enter logname: " user  
  
homedir=$(eval echo "$user")  
  
if [ ! -d "$homedir" ]; then  
    echo "User $user not found"  
    exit 1  
fi  
  
echo "Home directory of $user: $homedir"  
  
ls -l "$homedir"  
  
echo "Copying files/directories to /tmp..."  
cp -r "$homedir"/* /tmp/ 2>/dev/null  
  
echo "Current processes of $user:"  
ps -u "$user"
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/014_BashScript/Solutions/Lab2$ ./script.sh  
Enter logname: nagy  
Home directory of nagy: /home/nagy  
total 226932  
-rw-rw-r-- 1 nagy nagy      46205 Dec 14 21:55 aangpose013.jpg  
drwxrwxr-x 9 nagy nagy      4096 Dec 13 08:37 apache-tomcat-9.0.113  
drwxrwxr-x 2 nagy nagy      4096 Oct  9 20:02 app-backup  
drwxrwxr-x 3 nagy nagy      4096 Oct 10 07:37 build  
drwxrwxr-x 3 nagy nagy      4096 Jan  6 01:55 CascadeProjects  
drwxr-xr-x 9 nagy nagy      4096 Feb 19 11:05 Desktop  
-rw-rw-r-- 1 nagy nagy 108129318 Dec 15 20:06 discord.deb  
drwxr-xr-x 13 nagy nagy      4096 Feb 15 07:17 Documents  
drwxr-xr-x 7 nagy nagy      4096 Feb 22 12:34 Downloads  
drwxrwxr-x 2 nagy nagy      4096 Nov  3 18:33 drawio  
drwxrwxr-x 3 nagy nagy      4096 Aug 23  2025 eclipse
```
